

Auditing and Revalidating a Machine-Learning Diabetes Prediction System: A Case Study on Data Leakage and Clinical Deployment

Guillermo Álvarez Sánchez, Gabriel Iván Villafuerte Armenta,
Cristopher Yanhyu Muñoz Prado, Arianna Valentina Giannoccaro Quiñonez

Abstract—Machine-learning systems are increasingly embedded in clinical software, yet their reported performance is often accepted without methodological scrutiny. We report an audit of the diabetes-prediction module of PREDIA, a production electronic health-record platform whose model advertised a 97.89% accuracy. Working exclusively from the project’s own artifacts, we show that this figure is neither reproducible nor honest: the deployed estimator is a logistic regression hardcoded in TypeScript, trained on a different 947-sample dataset whose features (urea, creatinine, VLDL) are absent from the platform’s 100,000-row corpus, and its accuracy is inflated by *target leakage* through HbA1c, an official diagnostic criterion for diabetes. A five-fold cross-validated probe confirms that HbA1c alone attains 0.93 ROC-AUC, whereas a leakage-free *screening* model reaches only ≈ 0.66 . We retrain and compare nine classifiers under a leakage-controlled protocol with stratified cross-validation and a clinically oriented metric panel; logistic regression is the best honest model (ROC-AUC 0.659, MCC 0.223). We interpret it with permutation importance and SHAP, and migrate it into the PREDIA REST API as a self-contained TypeScript model with a redesigned, leakage-free feature contract. The study contributes a reproducible audit methodology and a candid account of the gap between inflated and honest metrics in clinical ML.

Index Terms—data leakage, diabetes prediction, model auditing, reproducibility, clinical decision support, interpretability, machine learning

I. INTRODUCTION

Diabetes mellitus is a leading global public-health burden, affecting an estimated 537 million adults worldwide [1]. Because early identification of at-risk individuals enables preventive intervention, machine learning (ML) has been widely proposed for diabetes risk stratification. However, the clinical value of such models depends entirely on the *validity* of their reported performance. When a model is evaluated incorrectly—most commonly through *data leakage*, the inadvertent use of information unavailable at prediction time or that encodes the outcome—reported metrics become severely inflated and the system fails silently in deployment [2], [3].

This paper is a case study of exactly such a failure, discovered while auditing PREDIA, a production electronic health-record (EHR) platform. PREDIA shipped a diabetes-prediction module advertising a 97.89% accuracy. Rather than accept this number, we subjected the model and its data to a reproducible audit using only the project’s own artifacts.

Objectives. (i) Determine whether PREDIA’s reported accuracy is valid and reproducible; (ii) test for data leakage and methodological flaws; (iii) establish an honest performance

baseline by retraining and comparing multiple classifiers under a leakage-controlled protocol; and (iv) deploy the selected model back into PREDIA without breaking the platform.

Contributions. (1) A formal, reproducible audit demonstrating that the deployed model’s accuracy is both irreproducible (trained on a different dataset with absent features) and inflated by HbA1c leakage. (2) A leakage-controlled benchmark of nine classifiers, separating an honest *screening* framing from a *clinical* framing that quantifies the leakage contribution (+0.285 ROC-AUC). (3) Interpretability and clinical analysis of the honest model. (4) A production migration of the revalidated model into the PREDIA REST API with a redesigned, leakage-free feature contract.

II. RELATED WORK

Tree- and kernel-based learners dominate tabular clinical prediction: random forests [4], gradient-boosted trees as implemented in XGBoost [5] and LightGBM [6], support-vector machines [7], and nearest-neighbour classifiers [8]. All experiments here use the `scikit-learn` ecosystem [9]. For model interpretability we adopt SHAP values [10] alongside permutation importance [4].

Data leakage is a long-recognised but persistent threat to the validity of ML studies [2]; recent work documents it as a primary driver of the reproducibility crisis in ML-based science across disciplines [3]. In clinical contexts, leakage frequently arises from including diagnostic measurements that *define* the label. The diagnosis of diabetes is governed by laboratory thresholds—HbA1c $\geq 6.5\%$ being a primary criterion [11]—so any model that consumes HbA1c to predict a HbA1c-derived label is, by construction, leaking. Because accuracy is misleading under class imbalance, we report the Matthews correlation coefficient (MCC), which is more informative for binary classification [12]. Finally, the original PREDIA model was trained on a small public diabetes dataset [13] whose feature set differs from the platform’s operational data—a discrepancy central to our findings.

III. METHODOLOGY

A. Dataset

All experiments use `diabetes_dataset.csv`, the corpus shipped with the PREDIA repository (Table I). It contains 100,000 instances and 31 attributes describing demographics, socio-economic status, lifestyle, anthropometry, vital signs,

TABLE I
CHARACTERISTICS OF THE `DIABETES_DATASET.CSV` CORPUS

Property	Value
Instances	100,000
Total columns	31
Numeric attributes	24 (incl. target)
Categorical attributes	7
Target variable	<code>diagnosed_diabetes</code>
Positive class (Diabetes)	59,998 (60.0%)
Negative class (No Diabetes)	40,002 (40.0%)
Imbalance ratio	1.50
Missing values	0
Duplicated rows	0
Train/test split	80,000 / 20,000 (stratified)

lipid profile and laboratory measurements. The binary target `diagnosed_diabetes` is mildly imbalanced (60.0% positive / 40.0% negative, ratio 1.50). The corpus has no missing values and no duplicated rows.

B. Leakage-aware feature taxonomy

The defining methodological decision of this study is an explicit feature taxonomy that controls leakage:

- **Direct leakage (always removed):** `diabetes_stage` (the diagnostic category itself) and `diabetes_risk_score` (a derived risk proxy).
- **Diagnostic laboratories:** `hba1c`, `glucose_fasting`, `glucose_postprandial` and `insulin_level`. Being diagnostic criteria, they are *excluded* from the honest *screening* framing and *included* only in a *clinical* framing used to quantify their (leaking) contribution.
- **Screening features:** demographics, socio-economic status, lifestyle, family history, anthropometry, vital signs and lipid profile—the 24 inputs that a pre-diagnostic risk tool can legitimately collect.

Fig. 1 confirms that the variables most correlated with the target are precisely the diagnostic laboratories, motivating this separation.

C. Preprocessing, validation and pipeline

Categorical attributes are one-hot encoded and numeric attributes standardised (*z*-score); binary attributes pass through. All transformations are fitted *inside* a `scikit-learn` Pipeline [9] to prevent train/test contamination. We use an 80/20 stratified split and 4-fold stratified cross-validation for model selection, with a global random seed of 42 for full reproducibility. Fig. 2 summarises the end-to-end protocol.

IV. AUDIT OF THE ORIGINAL MODEL

A. Architecture and reported performance

Inspection of the serialized artifact (`modelo_diabetes.pkl`) shows the deployed estimator is a *logistic regression* with 11 input features (Gender, AGE, Urea, Cr, HbA1c, Chol, TG, HDL, LDL, VLDL, BMI), a single intercept of 7.578, trained on 947

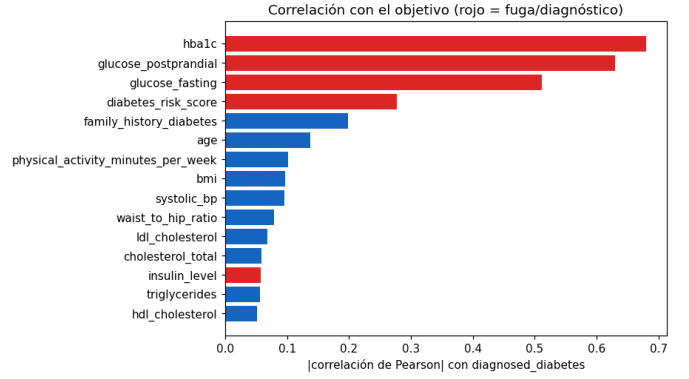


Fig. 1. Absolute Pearson correlation with the target. The strongest signals (red) are diagnostic/leaky variables (`hba1c`, `glucose_*`, `diabetes_risk_score`); legitimate screening variables (blue) are weakly correlated.

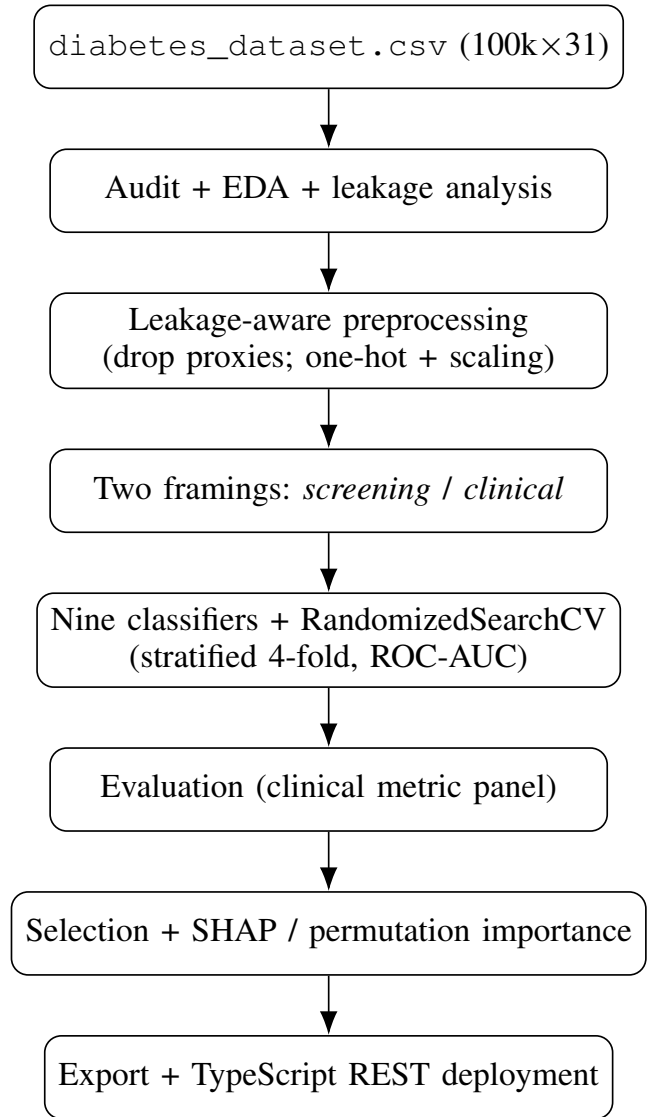


Fig. 2. Reproducible audit-to-deployment pipeline.

TABLE II
LEAKAGE DEMONSTRATION: 5-FOLD CROSS-VALIDATED LOGISTIC
REGRESSION TRAINED ON ISOLATED FEATURE SUBSETS OF
DIABETES_DATASET.CSV

Feature subset	Accuracy	ROC-AUC
hba1c only	0.8520	0.9325
glucose_fasting only	0.7307	0.8082
Diagnostic labs (HbA1c+glucose+insulin)	0.8567	0.9337
diabetes_stage+risk_score	0.9979	0.9974
Screening (non-diagnostic numeric)	0.6163	0.6121

samples (757 train / 190 test). The platform metadata and database seed report an accuracy of 0.9789, while the in-code documentation simultaneously states 98.42% accuracy, 100% precision and 0.9975 AUC—an internal inconsistency that already undermines confidence.

B. Reproducibility and dataset/feature mismatch

No training code or notebook exists in the repository; the methodology is therefore not reproducible. Moreover, the deployed model was trained on a different corpus (the 947-sample public dataset [13]): three of its eleven features—Urea, Cr and VLDL—are *absent* from PREDIA’s operational dataset, and the lipid features use an incompatible unit scale. The model cannot be evaluated as-is on the platform’s own data. We further observe that, at serving time, the model is not loaded from the .pkl file at all but reimplemented as hardcoded coefficients in TypeScript, augmented by hand-written clinical override rules.

C. Data leakage

HbA1c is the second most important feature of the original model, yet $\text{HbA1c} \geq 6.5\%$ is the diagnostic criterion for diabetes [11]. Using it as a predictor therefore leaks the label. Table II quantifies this with five-fold cross-validated logistic regressions on isolated feature subsets: HbA1c alone already yields 0.93 ROC-AUC, the pure-proxy pair `diabetes_stage+risk_score` reaches 0.997, whereas a leakage-free screening subset attains only 0.61. The reported $\approx 98\%$ accuracy is thus only attainable with diagnostic/leaking variables; it reflects leakage, not pre-diagnostic predictive skill.

V. EXPERIMENTATION

We retrained nine classifiers under the screening framing: logistic regression, random forest [4], extra trees, histogram gradient boosting, XGBoost [5], LightGBM [6], support-vector machine (RBF kernel) [7], k -nearest neighbours [8] and a multilayer perceptron. Each estimator is wrapped in the common preprocessing pipeline and tuned with `RandomizedSearchCV` (8 candidates, stratified 4-fold, scoring ROC-AUC). Because the kernel SVM and KNN scale poorly to 8×10^4 training rows, they are tuned on an 8,000-row stratified subsample and evaluated on the full test set; this is reported explicitly. All runs share `random_state=42`.

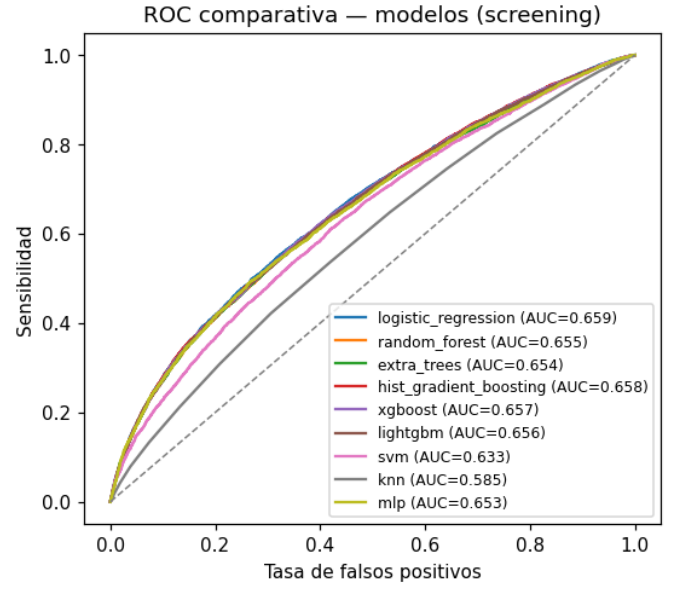


Fig. 3. Comparative ROC curves of the nine classifiers (screening framing).

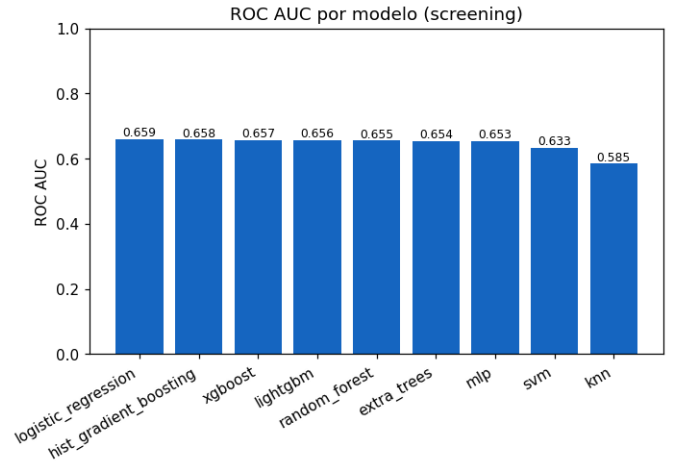


Fig. 4. ROC-AUC per model (screening framing).

VI. RESULTS

Table III reports the test-set performance of all nine models under the honest screening framing. Three findings stand out. First, the honest problem is genuinely hard: every model clusters around ROC-AUC 0.65–0.66. Second, *logistic regression is the best honest model* by ROC-AUC (0.6591) and by MCC (0.2229), and is also the best calibrated and most balanced. Third, the tree ensembles and SVM trade specificity for sensitivity (e.g. SVM reaches 0.96 sensitivity but 0.09 specificity), which inflates F1 without improving discrimination. The comparative ROC and ROC-AUC summaries are shown in Figs. 3 and 4; the confusion matrix and reliability diagram of the selected model appear in Figs. 5 and 6.

To quantify the leakage contribution directly, Table IV contrasts the screening framing with a clinical framing that re-

TABLE III

TEST-SET PERFORMANCE OF NINE CLASSIFIERS UNDER THE HONEST *screening* FRAMING (NO DIAGNOSTIC LABORATORY FEATURES). MODELS SORTED BY ROC-AUC. BEST VALUE PER COLUMN IN BOLD. † TUNED ON AN 8,000-ROW STRATIFIED SUBSAMPLE.

Model	Acc.	Bal. Acc.	Prec.	Sens.	Spec.	F1	ROC-AUC	MCC	CV-AUC
Logistic Regression	0.6044	0.6136	0.7144	0.5675	0.6597	0.6325	0.6591	0.2229	0.6610
Hist. Gradient Boosting	0.6310	0.5790	0.6489	0.8391	0.3189	0.7318	0.6579	0.1856	0.6591
XGBoost	0.6289	0.5781	0.6488	0.8317	0.3246	0.7289	0.6573	0.1817	0.6587
LightGBM	0.6280	0.5753	0.6464	0.8387	0.3119	0.7301	0.6558	0.1776	0.6574
Random Forest	0.6244	0.5594	0.6340	0.8848	0.2340	0.7387	0.6549	0.1576	0.6573
Extra Trees	0.6194	0.5438	0.6237	0.9220	0.1656	0.7441	0.6538	0.1356	0.6551
MLP (Neural Net)	0.6244	0.5821	0.6541	0.7937	0.3704	0.7172	0.6530	0.1807	0.6555
SVM (RBF) [†]	0.6123	0.5254	0.6130	0.9600	0.0907	0.7482	0.6328	0.1044	0.6360
KNN [†]	0.6003	0.5441	0.6268	0.8251	0.2631	0.7124	0.5845	0.1061	0.5872

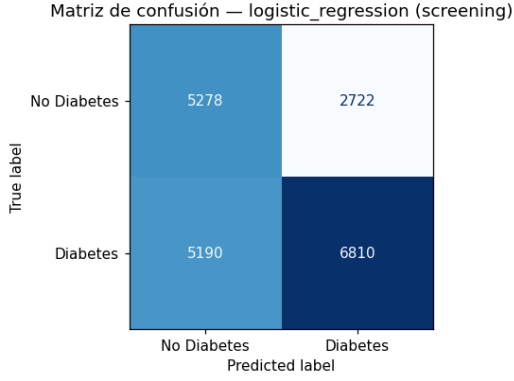


Fig. 5. Confusion matrix of the selected logistic-regression screening model on the 20,000-instance test set.

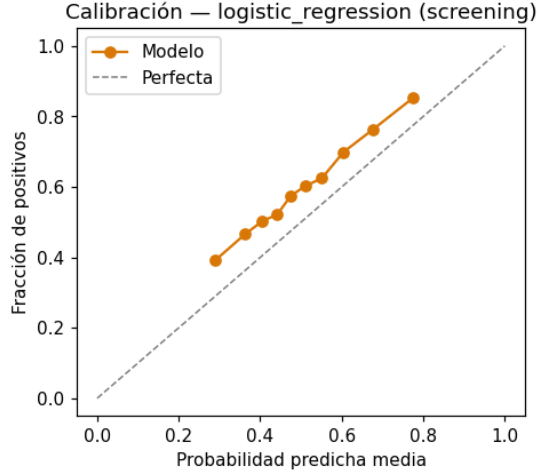


Fig. 6. Reliability (calibration) diagram of the selected model.

introduces the diagnostic laboratories. Adding those features raises ROC-AUC from ≈ 0.66 to ≈ 0.94 —a mean increase of $+0.285$ —confirming that the original system’s apparent excellence was almost entirely leakage.

TABLE IV

EFFECT OF INCLUDING DIAGNOSTIC LABORATORY FEATURES: HONEST *screening* FRAMING VS. *clinical* FRAMING (WITH HbA1c, GLUCOSE AND INSULIN). THE ROC-AUC GAP QUANTIFIES THE CONTRIBUTION OF INFORMATION LEAKAGE.

Model	Screening		Clinical	
	Acc.	ROC-AUC	Acc.	ROC-AUC
Logistic Regression	0.6044	0.6591	0.8878	0.9339
Random Forest	0.6244	0.6549	0.9196	0.9417
XGBoost	0.6289	0.6573	0.9199	0.9444
Mean Δ ROC-AUC	$+0.285$ (leakage contribution)			

TABLE V

MOST INFLUENTIAL FEATURES OF THE SELECTED SCREENING MODEL, BY PERMUTATION IMPORTANCE (DROP IN ROC-AUC) AND BY MEAN ABSOLUTE SHAP VALUE.

Permutation importance		SHAP (mean $ \phi $)	
Feature	Δ AUC	Feature	$ \phi $
family_history_diabetes	0.0816	age	0.0539
age	0.0340	family_history_diabetes	0.0536
physical_activity	0.0179	physical_activity	0.0362
bmi	0.0037	bmi	0.0220
diet_score	0.0023	diet_score	0.0126
hdl_cholesterol	0.0013	hdl_cholesterol	0.0125
screen_time	0.0012	triglycerides	0.0082
triglycerides	0.0008	screen_time	0.0076

VII. INTERPRETABILITY

We interpret the selected model with permutation importance (drop in test ROC-AUC) and SHAP values [10]. Both methods agree closely (Table V, Figs. 7 and 8): *family_history_diabetes*, *age* and *physical_activity_minutes_per_week* dominate, followed by *bmi* and *diet_score*.

Clinical impact. Reassuringly, the dominant predictors are *modifiable or routinely recordable* without laboratory testing (family history, physical activity, body-mass index, diet), which is exactly what a pre-diagnostic screening tool should

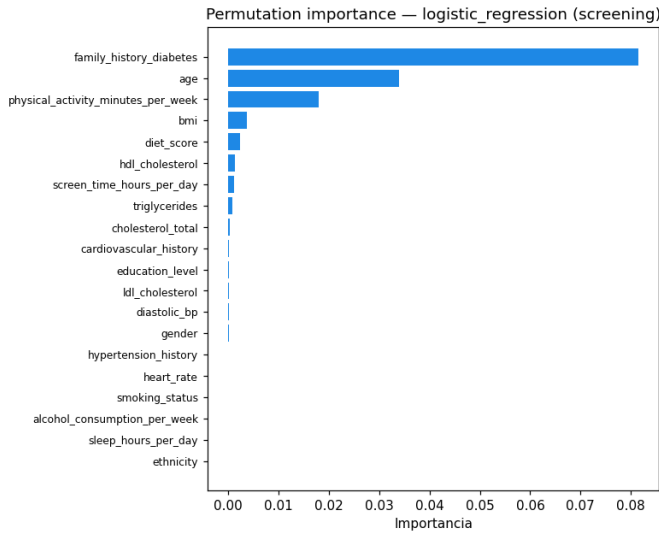


Fig. 7. Permutation importance of the selected screening model.

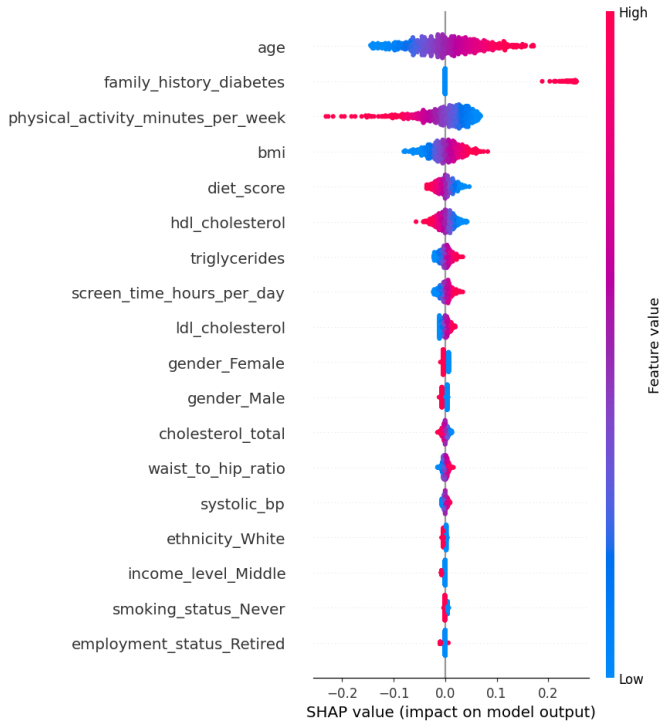


Fig. 8. SHAP summary (beeswarm) of the selected screening model.

rely upon. **Risks of interpretation.** Importance does not imply causation; the associations are observational, the corpus is of unknown provenance, and the modest effect sizes mean individual predictions carry substantial uncertainty.

VIII. DISCUSSION

Why the original model looked excellent. Its $\approx 98\%$ accuracy was the joint product of two artifacts: evaluation on a different, smaller dataset, and target leakage through HbA1c. Neither reflects the model’s ability to anticipate diabetes before

diagnosis. **Inflated vs. honest metrics.** The same algorithm (logistic regression) moves from ROC-AUC 0.93 (with leakage) to 0.66 (leakage-free); accuracy under imbalance is especially misleading here, which is why we emphasise MCC and balanced accuracy [12]. **Clinical implications.** An honest pre-diagnostic signal of ROC-AUC ≈ 0.66 is modest; the model is defensible only as decision *support* and must never be presented as a diagnosis. **Dataset limitations.** The corpus is synthetic-looking, of undocumented provenance, lacks external validity, and bundles diagnosis-defining variables that make naive modelling dangerous. **Deployment risks.** Shipping a leakage-inflated model invites automation bias and unsafe clinical reliance; this motivated both the revalidation and the redesigned feature contract.

IX. PRODUCTION INTEGRATION

The revalidated model was integrated into PREDIA while preserving the platform’s constraints. (i) **Architecture decision:** the serving runtime is pure TypeScript (no Python sidecar), chosen for portability; we therefore exported the logistic-regression screening model—which is also the best honest model—as plain coefficients plus the one-hot/scaling parameters, consumed generically at inference. (ii) **Export:** the full scikit-learn pipeline is serialized (joblib/pickle) for reproducibility, and a JSON parameter file is generated for the TypeScript runtime. (iii) **REST API:** the `/api/predicciones/nueva` endpoint and its validation schema were migrated to the new 24-feature, leakage-free contract; the web prediction form was rewritten accordingly. (iv) **Compatibility:** prediction *reads* (e.g. the mobile application) were unaffected, since only prediction *creation* changed; reported model metadata was corrected from the inflated 97.89% to the honest value. An end-to-end test confirmed the migrated endpoint returns coherent predictions.

X. CONCLUSIONS

We audited a production diabetes-prediction system and showed, using only the project’s own evidence, that its headline 97.89% accuracy was neither reproducible nor honest: it stemmed from a dataset/feature mismatch and from HbA1c-mediated target leakage. A leakage-controlled re-benchmark of nine models established an honest baseline (logistic regression, ROC-AUC 0.659, MCC 0.223), which we interpreted and redeployed into PREDIA with a corrected feature contract. The central lesson is methodological: in clinical ML, near-perfect metrics are a warning sign, and disciplined leakage control, reproducibility and honest reporting matter more than headline accuracy.

XI. FUTURE WORK

Future directions include: collecting a larger, clinically labelled and documented cohort; multicentric and temporal external validation; probability calibration and threshold optimisation against clinical cost (favouring sensitivity); subgroup fairness analysis across ethnicity and sex; privacy-preserving *federated learning* across institutions; and richer explainable-AI tooling integrated into the clinician workflow.

REPRODUCIBILITY

All artifacts (executed notebooks, trained models, metrics, figures and reports) are provided in the `ml-research/` directory with a pinned `requirements.txt` and a global seed of 42.

REFERENCES

- [1] International Diabetes Federation, “IDF diabetes atlas, 10th ed.” International Diabetes Federation, Brussels, Belgium, Tech. Rep., 2021.
- [2] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, “Leakage in data mining: Formulation, detection, and avoidance,” *ACM Transactions on Knowledge Discovery from Data (TKDD)*, vol. 6, no. 4, pp. 1–21, 2012, doi: 10.1145/2382577.2382579
- [3] S. Kapoor and A. Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” *Patterns*, vol. 4, no. 9, p. 100804, 2023, doi: 10.1016/j.patter.2023.100804
- [4] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324
- [5] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794, doi: 10.1145/2939672.2939785
- [6] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 3146–3154.
- [7] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995, doi: 10.1007/BF00994018
- [8] T. Cover and P. Hart, “Nearest neighbor pattern classification,” *IEEE Transactions on Information Theory*, vol. 13, no. 1, pp. 21–27, 1967, doi: 10.1109/TIT.1967.1053964
- [9] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [10] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.
- [11] American Diabetes Association, “Classification and diagnosis of diabetes: Standards of medical care in diabetes—2021,” *Diabetes Care*, vol. 44, no. Supplement 1, pp. S15–S33, 2021, doi: 10.2337/dc21-S002
- [12] D. Chicco and G. Jurman, “The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation,” *BMC Genomics*, vol. 21, no. 1, p. 6, 2020, doi: 10.1186/s12864-019-6413-7
- [13] A. Rashid, “Diabetes dataset,” Mendeley Data, V1, 2020, doi: 10.17632/wj9rwkp9c2.1